

Process Mining and Management 2025-2026

10 December 2025

Instructions

- Write clearly; illegible answers will not be marked.
- Take care to identify each answer clearly with:
 - the number of the exercise.
 - where appropriate, the part of the exercise you are answering.
- Clearly cross out unwanted answers before handing in your answers.
- You are not allowed to use lecture material.

BPMN Modeling and Analysis

Exercise 1 (BPMN Modeling).

Let us consider the following scenario for client request handling at a IT helpdesk:

*IT helpdesk request handling.*¹

The clients of the IT helpdesk are employees of a company. A request may be an IT-related problem that a client has, or an access request (e.g., requesting rights to access an IT system). Requests need to be handled according to their type and their priority.

A client makes a request to the help desk. The help desk staffed with “Level-1” support staff registers the request. Level-1 support staff typically are junior people with less than 12 months experience, but are capable of resolving simple requests. When the Level-1 employee knows how to resolve a request, she resolves it, notifies the resolution to the client and closes the simple request.

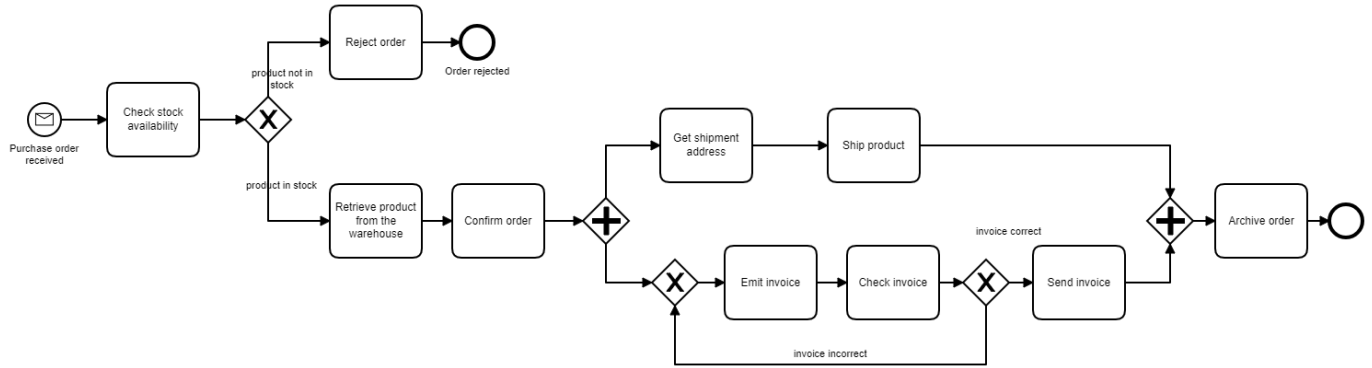
When the Level-1 employee does not know how to resolve a request, the request is forwarded to a more experienced “Level-2” support staff. When a Level-2 employee receives a request, the request is evaluated, a resolution is developed and sent back to the Level-1 employee. Eventually, the Level-1 employee forwards the resolution to the client who can test the resolution and notify the outcome to the IT helpdesk. The client notifies the outcome of the test to the Level-1 employee via e-mail. If the client states that the request is fixed, the request is closed successfully, and the process ends. If the request is not fixed, it is resent to Level-2 support for further action and goes through the process again. If, instead, after the client notification, no test outcome is received back for a month, the Level-1 employee closes the request for timeout and the process ends.

¹The described scenario is inspired from a scenario reported in the *Handbook of Business Process Management*, Vol. 1.

Model the scenario as a BPMN process model.

Exercise 2 (BPMN Understanding and analysis).

Let us consider the process model reported below:



and the cycle and processing time of the activities reported in the table below:

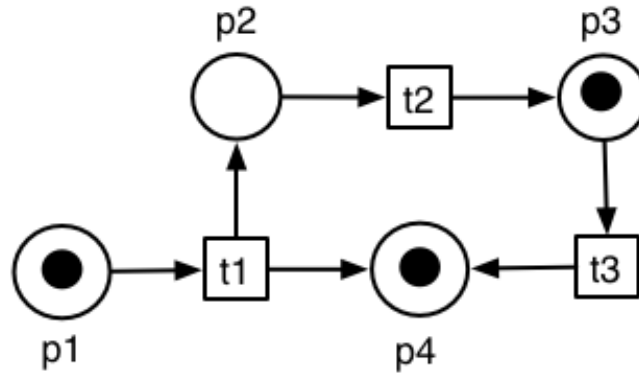
Activity	Cycle time	Processing time
Check stock availability	1 day	30 min
Reject order	7 days	1 hour
Retrieve product from the warehouse	3 days	4 hours
Confirm order	4 days	2 hours
Get shipment address	1 day	30 min
Ship product	10 days	24 hours
Emit invoice	1 day	1 hour
Check invoice	1 day	30 min
Send invoice	2 days	30 min
Archive order	4 days	1 hour

1. Describe the process trying to clearly define the BPMN constructs used for modeling details on the control flow, resources and data objects.
2. Classify the activities in this process into three categories: value adding (VA), business value-adding (BVA) and non-value-adding (NVA).
3. Assuming that in 20% of the cases the orders are rejected (because of the lack of the product in the warehouse) and that in 10% of the cases there is an issue with the invoice and that a day corresponds to a working day of 8 hours, compute cycle time (CT), processing time (PT) and cycle time efficiency (CTE) for this process.

Petri Nets

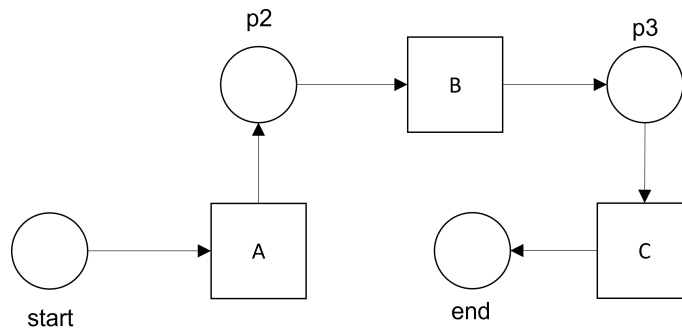
Exercise 3 (Basic Petri Nets).

Consider the Petri Net system depicted below.



1. Formalise the net as a 4-tuple $\langle P, T, F, m_0 \rangle$.
2. Give the preset and the postset of each transition.
3. Which transitions are enabled at m_0 ?
4. Give the list of all reachable markings by drawing the reachability graph corresponding to the Petri Net.
5. What are the reachable terminal markings? How do you identify them in the Transition System?
6. Is there a reachable marking where we have a non-deterministic choice?

Exercise 4 (Conformance checking). Consider the Petri Net system depicted below.



And trace $\langle A, B, B \rangle$. Replay the trace to check for conformance and

- Say whether it is conformant or not;
- Provide the numbers for p = produced tokens, c = consumed tokens, m = missing tokens, r = remaining tokens. In case of remaining or missing tokens say which are the problematic places (or draw the annotated Petri Net).

[Note: this is a simple net. The exercises in the exam may be slightly more complicated. Review the ones done during the lectures.]

Process Discovery

Exercise 5 (Heuristic Miner).

Let us consider the following event log extracted from some information system:

$$L = [\langle a, b, d \rangle^{100}, \\ \langle a, c, b, d \rangle^{80}, \\ \langle a, b, c, d \rangle^2, \\ \langle a, c, d \rangle^{40}, \\ \langle a, d \rangle^1]$$

1. Compute the frequency matrix $|>_L|$.
2. Compute the dependency measures matrix $|\implies_L|$.
3. Construct the dependency graph using a threshold of 0.90 for $|\implies_L|$, the threshold for direct succession ($|>_L|$) is 1.